

Healthcare Provider Fraud Detection Data Cleaning Process

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Updated: 2025-04-15

Dataset Source: Medicare Provider Fraud Detection on Kaggle

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1. Introduction

This document provides comprehensive technical documentation for the data cleaning and preparation phase of the Healthcare Provider Fraud Detection project. The goal was to clean and structure eight interrelated datasets so they are ready for analysis and machine learning modeling. This includes handling missing data, type conversions, feature engineering, referential integrity validation, and outlier control without compromising row counts.

2. Loading Data

The following eight datasets were loaded from raw CSV files using pandas:

- Train and Test Beneficiary Data
- Train and Test Inpatient Claims
- Train and Test Outpatient Claims
- Train and Test Feature Tables

Each file was read using `pd.read_csv()` and stored in separate DataFrames.

3. Handling Missing Values

Missing values were handled with different strategies based on data type:

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- Dates: Parsed with `errors='coerce'`, missing values become `NaT`
- Numerical Columns: Imputed using median
- Categorical/Object Columns: Imputed using the mode (most frequent value)
- Columns with all missing values were preserved for structure consistency.

```
df['DOB'] = pd.to_datetime(df['DOB'], errors='coerce')
df['DOD'] = pd.to_datetime(df['DOD'], errors='coerce')
df[numerical_cols] = df[numerical_cols].fillna(df[numerical_cols].median())
df[categorical_cols] = df[categorical_cols].fillna(df[categorical_cols].mode().iloc[0])
```

4. Handling Outliers

Outliers were initially removed using row-wise IQR filtering, which caused significant row loss. This was replaced by a winsorization approach, which caps extreme values while preserving dataset size. Only columns with sufficient variance and more than 1000 non-null values were considered for winsorization.

```
def winsorize_column(series):
    Q1 = series.quantile(0.25)
    Q3 = series.quantile(0.75)
    IQR = Q3 - Q1
    return series.clip(Q1 - 1.5*IQR, Q3 + 1.5*IQR)
```

5. General Data Cleaning

```
def data_cleaning_general(df):
    for col in df.select_dtypes(include=['float64', 'int64']).columns:
        if df[col].notnull().sum() > 1000 and df[col].std() > 0:
            df[col] = winsorize_column(df[col])
    return df
```

6. Data Type Conversion

```
def convert_date_columns(df, date_columns):
    for col in date_columns:
        df[col] = pd.to_datetime(df[col], errors='coerce')
    return df

def convert_to_category(df, categorical_columns):
    df[categorical_columns] = df[categorical_columns].astype('category')
    return df
```

7. Applying Data Cleaning

```
train_beneficiary = add_engineered_features(data_cleaning_beneficiary(train_beneficiary))
...
```

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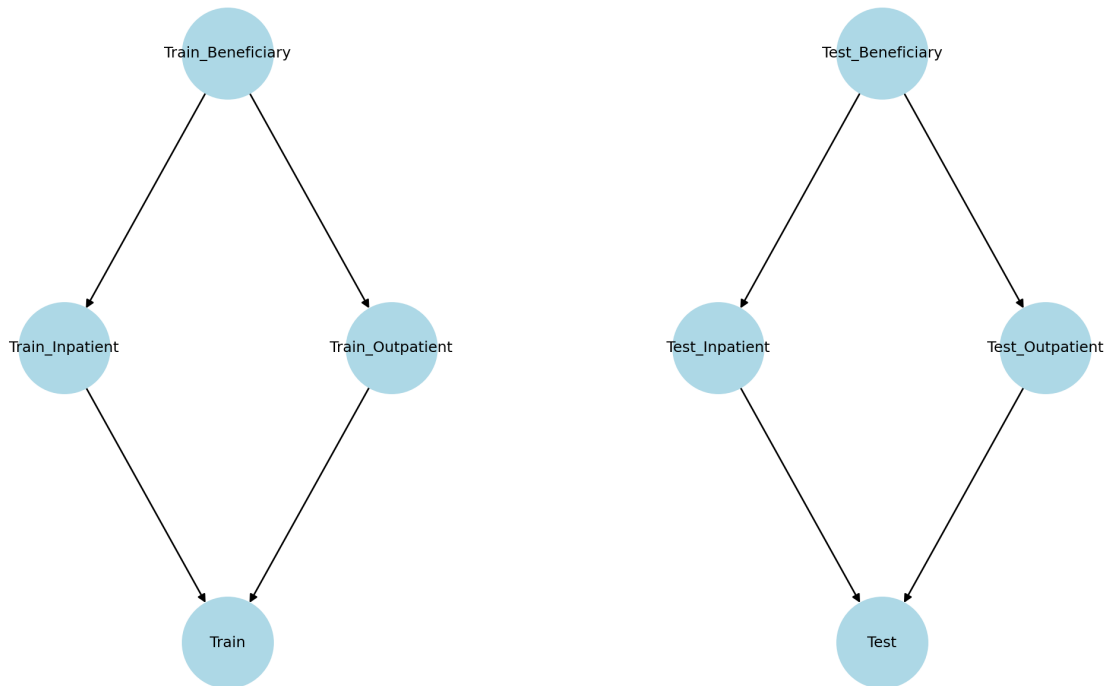
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```
train_outpatient = convert_to_category(train_outpatient, categorical_columns_outpatient)
```

8. Dataset Relationship Diagram

Diagram showing relationships among datasets based on BeneID:

Dataset Relationships via BeneID



9. Verification

- No BeneID mismatches found
- Row counts preserved after cleaning
- Train/Test datasets aligned

10. Challenges and Lessons Learned

- Applying row-wise outlier removal resulted in nearly complete data loss due to aggressive filtering. This was corrected by using winsorization.
- Many columns were sparse (e.g. ClmProcedureCode_6), but were kept for consistency.
- Referential integrity checks revealed no mismatches, validating the linkage structure.

What was learned:

- Always preview how filtering affects row count before applying at scale
- Winsorization is safer than deletion when preserving training size is important
- Not all columns with missing values are problematic - some are informative by presence/absence

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11. Recommendations

- Drop or combine sparse features before modeling
- Create new features like total claims per provider, or diagnosis code frequency
- One-hot encode or label encode categoricals
- Apply class balancing or reweighting if PotentialFraud is imbalanced

12. Conclusion

All raw datasets have been cleaned, transformed, and validated. Outlier control is now non-destructive, types are correctly assigned, missing values handled, and the data is structurally aligned across train/test. The project is ready for EDA, modeling, or deployment pipelines.

13. Detailed Process Overview

- Cleaned 8 raw datasets using pandas
- Filled missing dates, numerics, categoricals with appropriate logic
- Converted all date and categorical fields consistently
- Applied winsorization to numerical columns with enough data
- Added engineered feature AgeAtDeathOrLastClaim
- Verified dataset alignment and referential integrity
- Preserved row counts while removing numeric skew
- Used `.info()` and `.isnull().sum()` for structural validation